III. Minkowski Thoory & its applications : ① The finitmess of the Class Number
② Dirichlet unit theorems

1. Minkowski Theory.

Def III.1.1. V be the vector space $\overset{\text{over } \mathbb{R}}{,}$ $\dim V = n$. A lattice Λ in V is a subgroup of the form

$\Lambda = \mathbb{Z}e_1 + \dots + \mathbb{Z}e_r$. e_i linearly independent

When $r = n$, the lattice is said to be full.

Notation III.1.2 Λ be full. $\operatorname{covol}(\Lambda) := \operatorname{vol}(P)$ where $P = \{\sum_{i=1}^{n} t_i v_i \mid 0 \le t_i < 1\}$.

Some estimation & caculus.

Thm III.1.3. $\Lambda \subseteq V$ full, $S \subseteq V$ measurable. If $\operatorname{vol}(S) > \operatorname{coivol}(\Lambda)$
Then $\exists\, x \neq y \in S$. s.t. $x - y \in \Lambda$.

Thm III.1.4. $\Lambda \subseteq \mathbb{R}^n$ full, $C \subseteq \mathbb{R}^n$ compact. convex. symmetric of the origin.
If $\operatorname{vol}(C) \ge 2^n \operatorname{covol}(\Lambda)$, Then $C \cap (\Lambda \setminus \{0\}) \neq \phi$.

Def III.1.5. $\mathfrak{a} \subseteq \mathcal{O}_K$. Define the norm of $\mathfrak{a}$. $N\mathfrak{a} := [\mathcal{O}_K : \mathfrak{a}] = |\mathcal{O}_K / \mathfrak{a}|$

Prop. III.1.6. $\forall \mathfrak{a}, \mathfrak{b} \subseteq \mathcal{O}_K$. $N(\mathfrak{a}\mathfrak{b}) = N(\mathfrak{a})N(\mathfrak{b})$

Thm III.1.7. If $0 \neq \alpha \in \mathcal{O}_K$, then $N((\alpha)) = |Nm_{K/\mathbb{Q}}(\alpha)|$

Def III.1.8. K be the $\overset{\text{algebraic}}{\text{number}}$ field has r real-embeddings, $2s$ complex-embeddings
$\forall x \in K$ Define $\sigma(x) = (\sigma_1(x), \sigma_2(x), \dots, \sigma_{r+1}(x), \dots, \sigma_{r+s}(x))$, V be $\mathbb{R}^r \times \mathbb{C}^s$
The norm of x $N(x) = x_1 \cdots x_r |z_{r+1}|^2 \cdots |z_{r+s}|^2$

Rmk. By computation, we have $N(\sigma(\alpha)) = Nm_{K/\mathbb{Q}}(\alpha)$.

Prop. III.1.9. $\mathfrak{a} \subseteq \mathcal{O}_K$ Then $\sigma(\mathfrak{a})$ is a full lattice in V. And we have $\operatorname{covol}(\sigma(\mathfrak{a})) = 2^{-s} N\mathfrak{a} \sqrt{|\Delta_K|}$
where $|\Delta_K| = D(\omega_1, \dots, \omega_n)$ ω_i be the integral basis of $\mathcal{O}_K$.

proof:

Def Ⅲ.1.10. $\forall t>0$, Let $X(t)=\{x\in V \mid \sum_{i=1}^{r}|x_i|+2\sum_{j=r+1}^{r+s}|z_j|\le t\}$.

Lemma Ⅲ.1.11. Let $n=r+2s$. Then $\mathrm{vol}(X(t))=2^r\cdot(\frac{\pi}{2})^s\frac{t^n}{n!}$

pf:

2. ⓠ Finiteness of Class number, Dirichlet unit theorem.

Prop. Ⅲ 2.1. Let $0\neq\mathfrak{a}\subseteq\mathcal{O}_K$. Then $\exists\, 0\neq\alpha\in\mathfrak{a}$. s.t. $|N_{m\,K/\mathbb{Q}}(\alpha)|\le\frac{n!}{n^n}\cdot(\frac{4}{\pi})^s N\mathfrak{a}\sqrt{|\Delta_K|}$

proof:

Thm Ⅲ.2.2. (Minkowski) Let $B_K=\frac{n!}{n^n}(\frac{4}{\pi})^s\sqrt{|\Delta_K|}$. Then $\forall$ ideal class of K. $\exists\,\mathfrak{a}$ in it. such that $N\mathfrak{a}\le B_K$

proof:

Thm Ⅲ.2.3. K be the algebraic number field. Then $h_K = |Cl(K)|$ is finite.

proof: $\forall \mathfrak{a} \subseteq \mathcal{O}_K$. $\mathfrak{a} = \prod \mathfrak{p}_i^{e_i}$. The $N\mathfrak{a} = \prod (N\mathfrak{p}_i)^{e_i}$ is bounded

$$\forall q \in \mathbb{Z} \quad N(\mathfrak{a}) = q \Rightarrow |\mathcal{O}_K/\mathfrak{a}| = q \Rightarrow \bar{q} = \bar{o} \in \mathcal{O}_K/\mathfrak{a} \Rightarrow \mathfrak{a} \mid q\mathcal{O}_K.$$

Then we know that $\mathfrak{a}$ satisfying $N(\mathfrak{a}) = q$ is finite.

By Thm Ⅲ.2.2. Every class has an integral ideal $\mathfrak{b}$ s.t. $N\mathfrak{b}$ is bounded.

Thus the number of $\mathfrak{b}$ is finite. Thus h_K is finite $\square$.